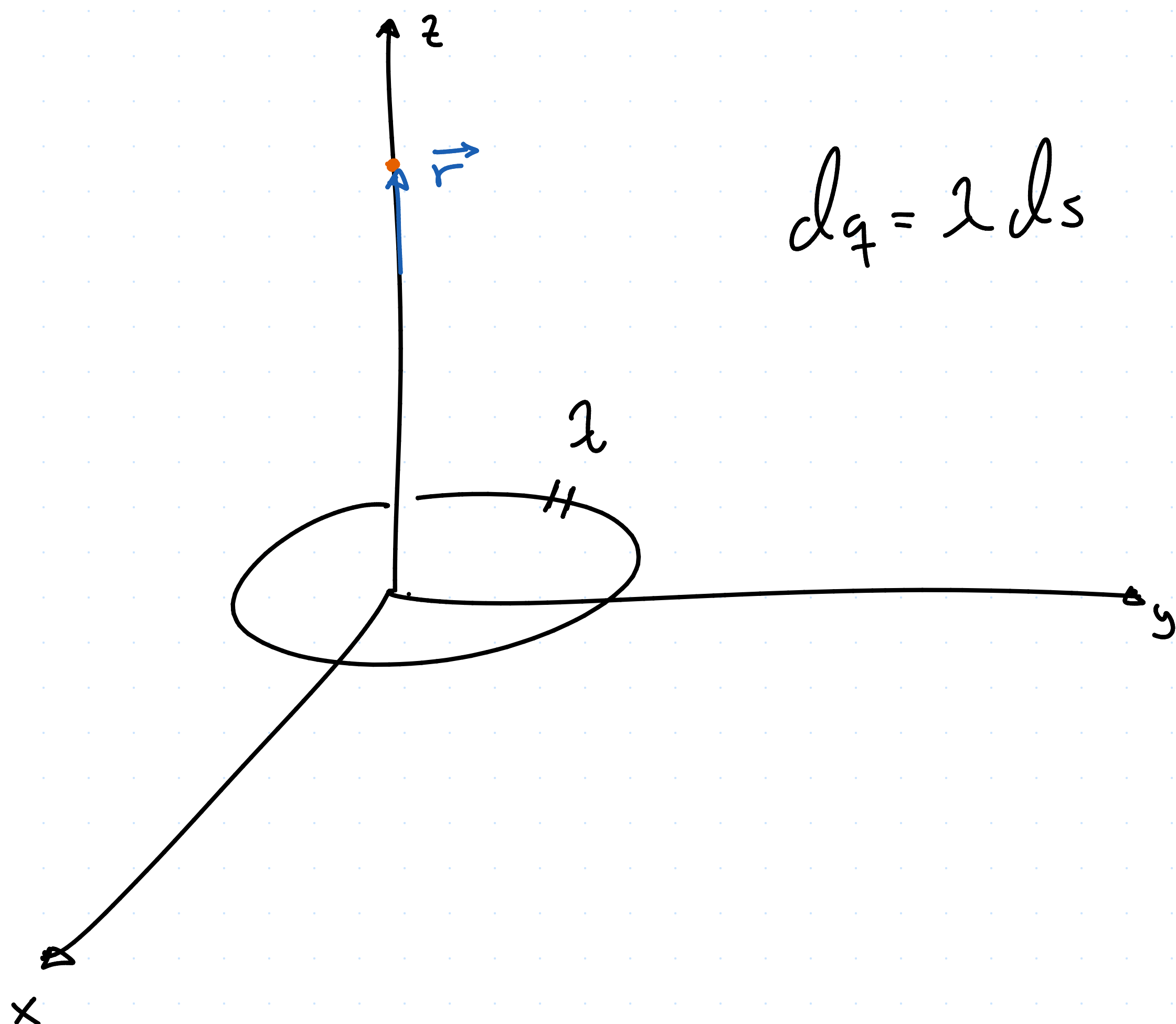




$$dq = \lambda ds \Rightarrow d\tau = \lambda R d\theta$$

$$\vec{r}' = R \cos \theta \hat{i} + R \sin \theta \hat{j} ; \quad \vec{r} = z \hat{k} ; \quad \vec{r} - \vec{r}'$$

$$= (-R \cos \theta, -R \sin \theta, z)$$

$$\therefore \|\vec{r} - \vec{r}'\| = (R^2 + z^2)^{1/2}$$

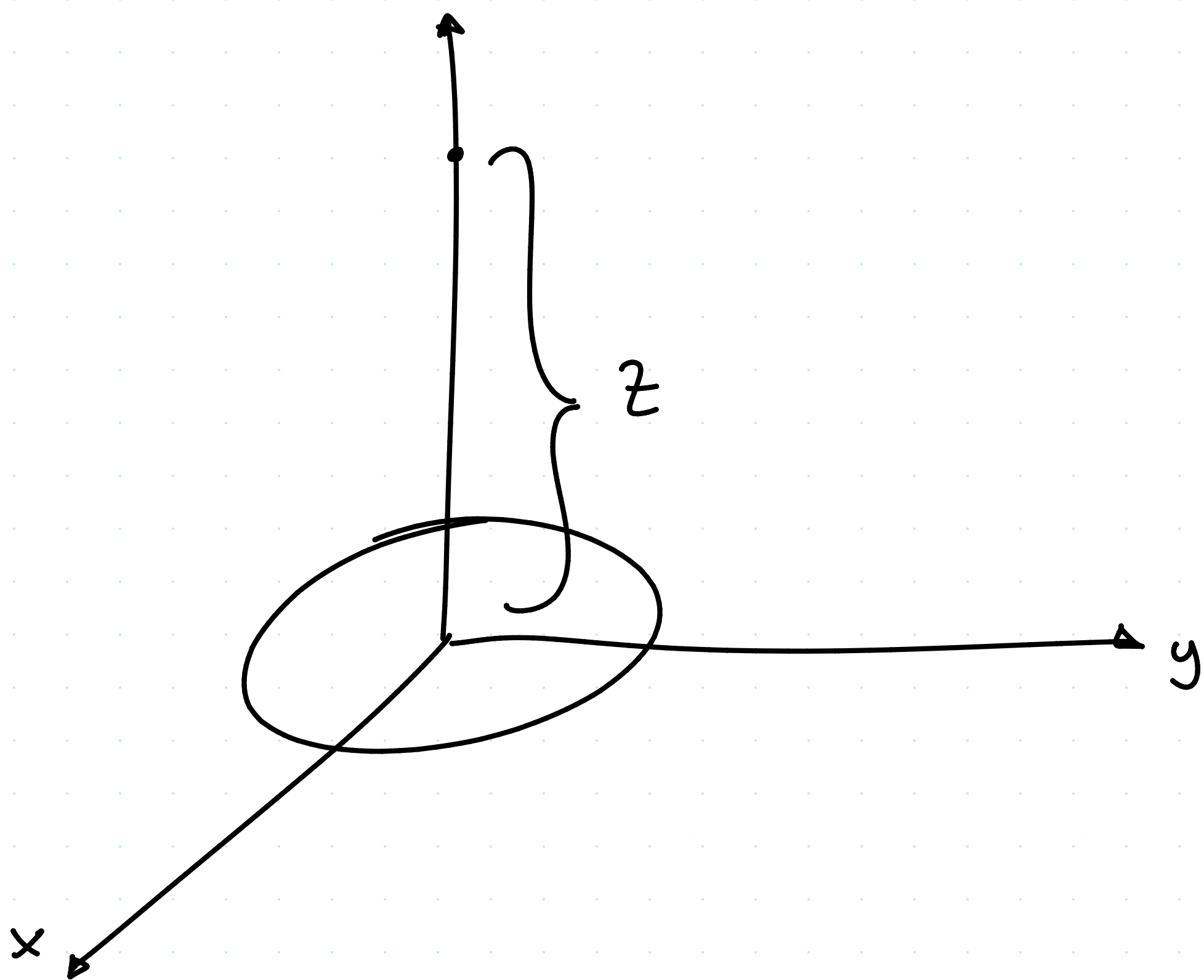
$$\Rightarrow \vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\lambda R d\theta}{(R^2 + z^2)^{3/2}} (-R \cos \theta \hat{i} - R \sin \theta \hat{j} + z \hat{k})$$

$$\vec{E} = -\nabla U$$

Subir esta tarea a
teams.

$$3) \vec{E} = \frac{\sigma}{2\epsilon_0} \hat{k} \quad \text{quando } r \rightarrow \infty$$

1)



Tenemos

$$U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_C \frac{\lambda R d\theta}{|\vec{r} - \vec{r}'|}$$

$$= \frac{\lambda R}{4\pi\epsilon_0} \frac{1}{|\vec{r} - \vec{r}'|} \int_C d\theta$$

$$(R^2 + z^2)^{1/2}$$

$$= \frac{\lambda R}{4\pi\epsilon_0} \left(\frac{1}{|\vec{r} - \vec{r}'|} \right) \int_0^{2\pi} d\theta$$

$$= \frac{\lambda R (2\pi)}{4\pi\epsilon_0 (R^2 + z^2)^{1/2}}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{Q}{\sqrt{R^2 + z^2}}$$

Calculamos el gradiente de $\nabla U(\vec{r})$ para hallar $\vec{E}$.

$$\vec{E} = -\nabla U(\vec{r}) ; \quad \frac{\partial U}{\partial x} = 0 ; \quad \frac{\partial U}{\partial y} = 0$$

Tenemos... $\frac{\partial U}{\partial x_i} = 0 \quad \{i\}_{i=1}^2$

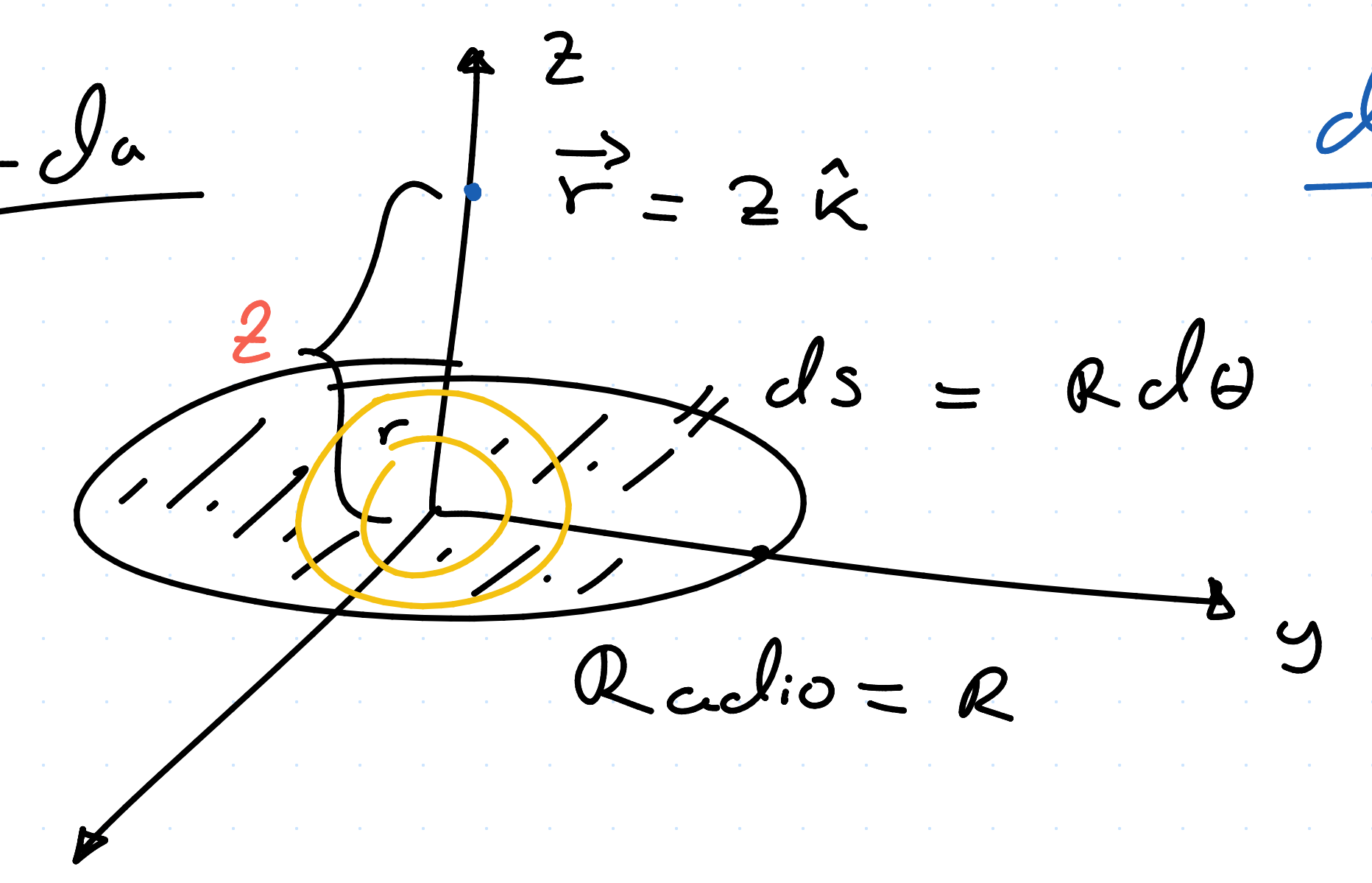
$$\begin{aligned} \frac{\partial U}{\partial z} &= \frac{Q}{4\pi\epsilon_0} \frac{\partial}{\partial z} (R^2 + z^2)^{-1/2} \\ &= \frac{Q}{4\pi\epsilon_0} \left(-\frac{1}{2}\right) \frac{2z}{(R^2 + z^2)^{3/2}} = -\frac{Q}{4\pi\epsilon_0} \left(\frac{z}{(R^2 + z^2)^{3/2}}\right) \end{aligned}$$

$$\Rightarrow \vec{E} = -\nabla U = \frac{Q}{4\pi\epsilon_0} \frac{z}{(R^2 + z^2)^{3/2}} \hat{k}$$

• Potencial del disco en el eje z .

$\vec{E} = \frac{1}{4\pi\epsilon_0} \int_S \frac{\sigma d\vec{a}}{r^2}$

$d\vec{a} = \sigma d\vec{a}$



$\vec{r} = z \hat{k}$

$ds = R d\theta$

$R_{radio} = R$

$\vec{r}' = R \cos \theta \hat{i} + R \sin \theta \hat{j}$

$U = r^2 + z^2$

$du = 2r dr$

$$\Rightarrow U(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_0^R \frac{\sigma (2\pi r dr)}{\sqrt{r^2 + z^2}}$$

$$\hookrightarrow \frac{dV}{2} = r dr$$

$$\Rightarrow V(\vec{r}) = \frac{2\pi\sigma}{4\pi\epsilon_0} \left(\frac{1}{2}\right) \int_0^R \frac{dV}{V^{1/2}}$$

$$= \frac{2\pi\sigma}{4\pi\epsilon_0} \left(\frac{1}{2}\right) \int_0^R V^{-1/2} = [\dots] \left[\frac{V^{-\frac{1}{2} + \frac{2}{2}}}{\frac{1}{2}} \right]$$

$$= [\dots] 2 V^{1/2} \Big|_0^R$$

$$\frac{\sigma}{4\epsilon_0} (z) (r^2 + z^2)^{1/2} \Big|_0^R$$

$$= \frac{\sigma}{2\epsilon_0} \left[(R^2 + z^2)^{1/2} - (z^2)^{1/2} \right] = \frac{\sigma}{2\epsilon_0} \left[(R^2 + z^2)^{1/2} - z \right]$$

$\vdots$

$$E = -\nabla V(\vec{r}) \rightarrow \frac{\partial V}{\partial x_i} = 0$$

$$\frac{\partial V}{\partial z} = \frac{\sigma}{2\epsilon_0} \frac{\partial}{\partial z} \left([R^2 + z^2]^{1/2} - z \right)$$

$$= \frac{\sigma}{2\epsilon_0} \frac{\partial}{\partial z} (R^2 + z^2)^{1/2} - 1$$

$$K = \frac{1}{4\pi\epsilon_0}$$

$$= \frac{\sigma}{2\epsilon_0} \left[\frac{1}{2} (R^2 + z^2)^{-\frac{1}{2}} (2z) - 1 \right]$$

$$E = -\nabla U(\vec{r})$$

$$= \frac{\sigma}{2\epsilon_0} \left[\frac{z}{\sqrt{R^2 + z^2}} - 1 \right]$$

$\Rightarrow$

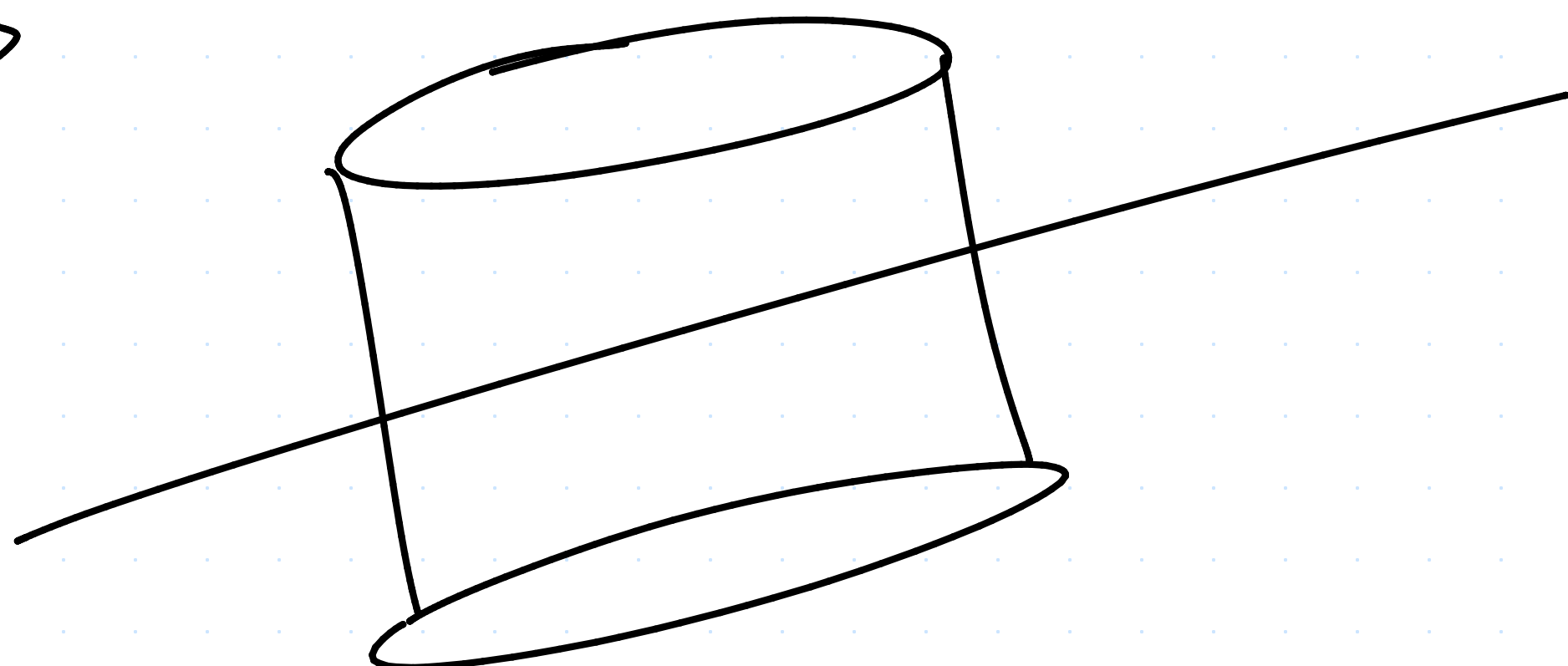
$$\frac{\sigma}{2\epsilon_0} \left[1 - \frac{z}{\sqrt{R^2 + z^2}} \right]$$

obtener campo por el otro método (analítico).

Si $R \rightarrow \infty \Rightarrow$

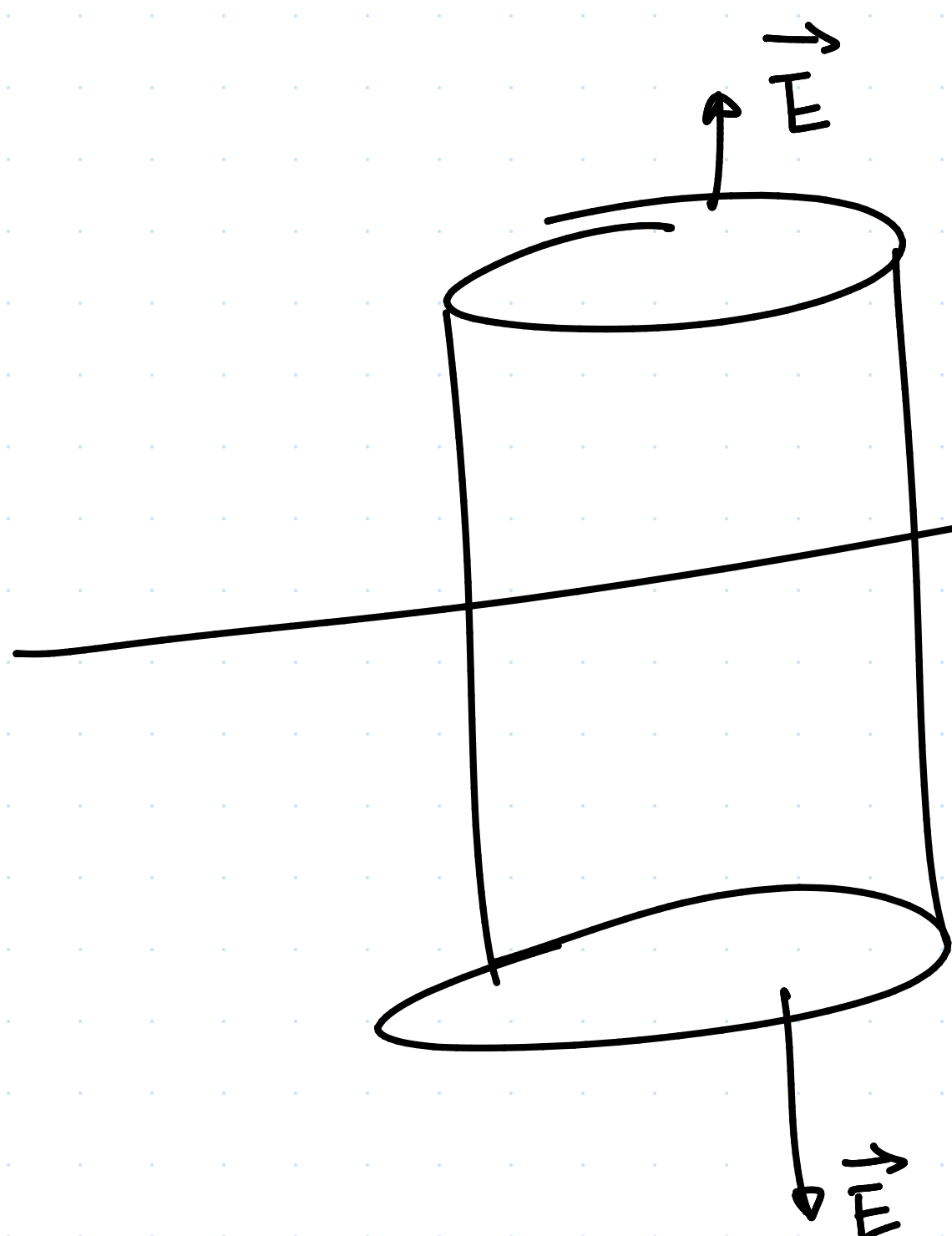
carga de tabletas

$$\vec{E} = \frac{2\pi\sigma}{4\pi\epsilon_0} = \frac{\sigma}{2\epsilon_0}$$



$$\oint_S \vec{E} \cdot d\vec{a} = \frac{q_{enc}}{\epsilon_0}$$

Tenemos



Área transversal (A)

contribución por ambas caras

$$\oint \vec{E} \cdot d\vec{a}$$

$$q_{enc} = \sigma A$$

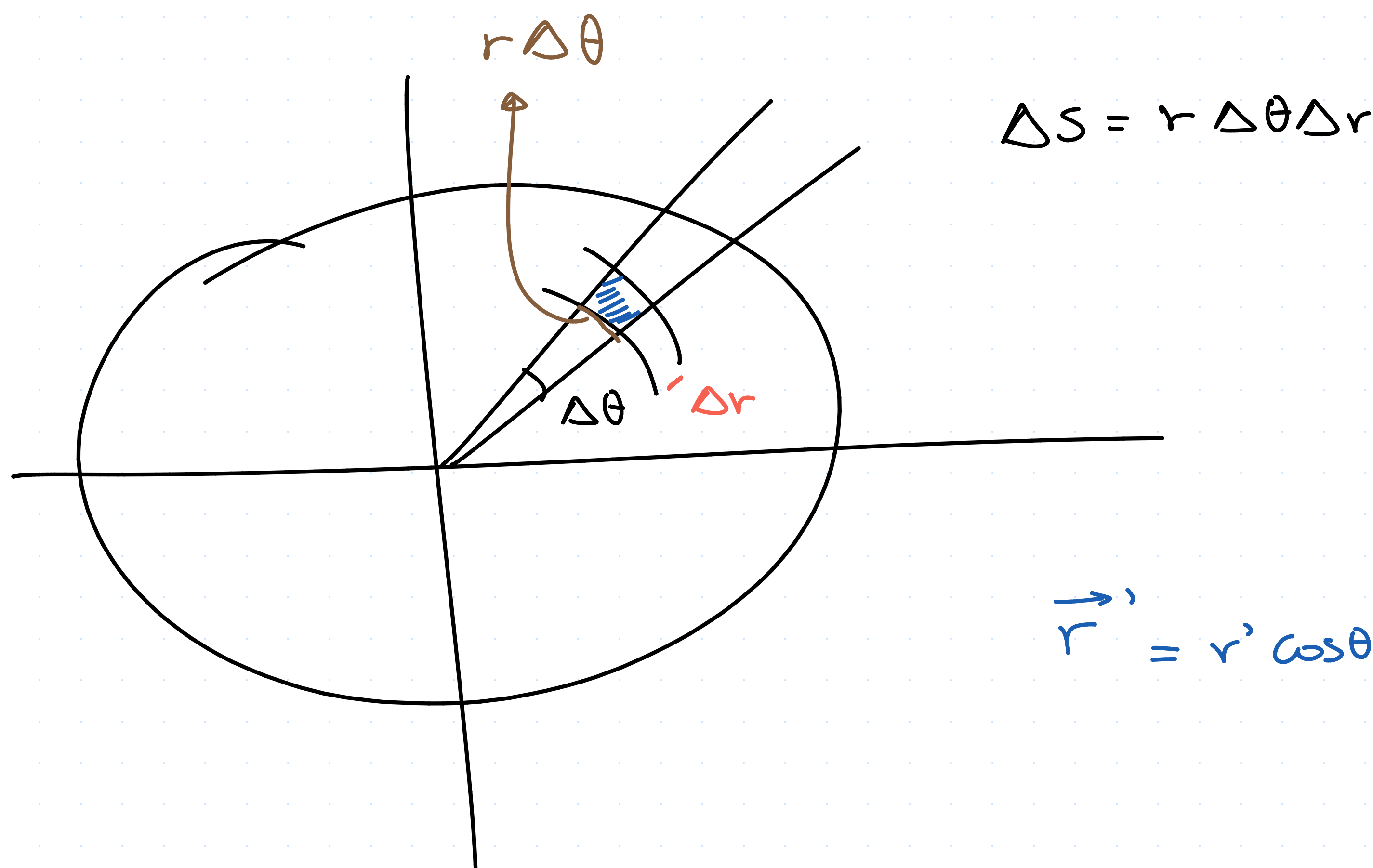
2 caras

$$= 2EA = \frac{\sigma A}{\epsilon_0}$$

$$\rightarrow \boxed{E = \frac{\sigma}{2\epsilon_0}}$$

Tenemos

$$V(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^R \frac{\sigma r d\theta dr}{\sqrt{r'^2 + z^2}}$$



$$\vec{r}' = r' \cos\theta \hat{i} + r' \sin\theta \hat{j}$$

$\vec{E} = -\nabla V(\vec{r}) \rightarrow$ Campo eléctrico es tangencial a la superficie de nivel (superficie equipotencial).

1) why $\vec{E} = 0$ inside the conductor (static) ?

2) why the extra charge in a conductor is on the surface of the conductor ?

3) why the surface of the conductor is a equipotential surface.

4) why $\vec{E}$ is perpendicular to the conductor's surface?